GLASS · METAL · CERAMIC

Laboratory Equipment, Instruments & Accessories

Shaping Materials for Science

GLASS • METAL • CERAMIC

Shaping materials for science

PRODUCT CATALOGUE 2026

OUR PRODUCT RANGE

- GENERAL GLASS APPARATUS
- CUSTOMIZED GLASS APPARATUS
- PHOTO REACTOR / BIO REACTOR
- SOLVENT PURIFICATION SYSTEM
- AUTO CLAVE
- HIGH-PRESSURE REACTOR
- HIGH-PRESSURE HYDROGEN DISTRIBUTION SYSTEM

QUALITY ASSURANCE | CUSTOMIZED SOLUTIONS | PRECISION ENGINEERED | DEPENDABLE PERFORMANCE

Vision and mission

- To become a leading provider of advanced glass, metal, and ceramic technologies that support scientific research, innovation, and industrial development.

- To design and manufacture high-quality laboratory equipment and customized scientific systems using advanced glass, metal, and ceramic materials.

- To continuously improve product quality, safety, and performance through engineering excellence and scientific expertise.

High Pressure Reactor/Autoclave

GMC Technology

High-Pressure Reactor
(Model No: HPR-GMC-01)

The **High-Pressure Reactor** is designed for laboratory-scale synthesis under elevated pressure and temperature conditions. The system consists of a stainless-steel **high-pressure reactor vessel** and a **separate digital temperature control unit**, providing precise heating and process control.

Its robust construction ensures reliable operation while providing excellent corrosion resistance and long service life. The precise control of pressure and temperature enables reproducible reaction conditions, making the system suitable for research, process development, and pilot-scale studies. The reactor is suitable for hydrogenation, catalytic reactions, hydrothermal synthesis, and other high-pressure chemical processes.

Available in various capacities to meet specific application requirements.

Technical Features

Volume	**50 ml to 100 ml (As per requirement)**
Pressure	**Up to**
Temperature	**Up to**
Material	
Magnetic Stirring	
Double sealed	**O-ring and medium seal metal to metal**

https://amarequip.com/pressure-reactor-auxiliaries-products/stirred-pressure-reactors-autoclaves-100-ml-to-100-l

High-Pressure Reactor
(Model No: HPR-GMC-02)

The **High-Pressure Reactor** is a compact, integrated system designed for carrying out laboratory-scale chemical reactions under controlled pressure and temperature conditions. The unit combines a stainless-steel pressure vessel, electric heating system, and digital temperature controller in a single assembly, enabling precise process control and safe operation. Equipped with a pressure gauge and high-pressure fittings, the reactor is suitable for hydrogenation, catalytic synthesis, hydrothermal reactions, polymerization, and other pressure-assisted chemical processes.

***Available in various capacities to meet specific application requirements*.**

Technical Features

Volume	**25 ml to 100 ml (As per requirement)**
Pressure	**Up to**
Temperature	**Up to**
Material	
Magnetic Stirring	
Double sealed	**O-ring and medium seal metal to metal**

https://www.premex-reactor.com/en/product/laborreaktoren/hochdruckreaktor-varioso/

High-Pressure Reactor
(Model No: HPR-GMC-03)

The **High-Pressure Stirred Reactor** is a compact bench-top system designed for laboratory-scale reactions under controlled pressure and temperature conditions. The reactor integrates a stainless-steel pressure vessel, magnetic stirring system, electric heating oven, and digital temperature controller into a single unit, ensuring precise control and reproducible process conditions. The modular design is suitable for research, process development, and pressure-assisted synthesis.

Available in various capacities to meet specific application requirements.

Technical Features

Volume	**50 ml to 100 ml (As per requirement)**
Pressure	**Up to**
Temperature	**Up to**
Material	
Magnetic Stirring	
Double sealed	**O-ring and medium seal metal to metal**

https://techinstro.com/high-pressure-autoclave-https://techinstro.com/high-pressure-autoclave-reactor/?srsltid=AfmBOop6fE0bxj4vyaGEJ5up281utOMB5QpZwzAggx-2CnJYWY_ef2OQ

High-Pressure Reactor-Autoclave

The **high-pressure autoclave** is a robust reaction vessel designed for conducting pressure-assisted chemical processes. The assembly comprises a stainless-steel reactor body, a pressure gauge, a pressure control valve, and high-pressure fittings to ensure accurate pressure monitoring and secure operation. It is suitable for research and process development involving high-pressure synthesis and catalytic reactions. These reactors enable safe and efficient execution of complex chemical processes such as hydrogenation, oxidation, polymerization, nitration, and catalytic reactions.

Technical features

Volume	
Pressure	**Up to**
Temperature	**Up to**
Material	

Hydrothermal Autoclave Reactor

The **Hydrothermal Autoclave Reactor** is designed for carrying out hydrothermal and solvothermal synthesis under elevated temperature and pressure conditions. The reactor consists of a high-strength material outer body with a PTFE-lined reaction chamber, providing excellent corrosion resistance and leak-proof performance. It is available in both Nut & Bolt and Threaded sealing designs with a range of standard reactor volumes to meet diverse laboratory and research requirements. The reactor is ideal for crystal growth, nanomaterial synthesis, catalyst preparation, material digestion, and other high-pressure chemical processes.

- ***Available Standard Specifications are 10ml, 25ml, 50ml, 100ml and customized according to user needs.***

Technical features

Maximum Temperature	
Maximum working pressure	
Material	
Sealing Design	**Nut & Bolt, Threaded**

Stirred-Tank BioReactor

The **Stirred-Tank Bioreactor** is designed for laboratory-scale microbial fermentation under precisely controlled operating conditions. The system is equipped with a top-mounted motor-driven impeller to provide efficient mixing, uniform nutrient distribution, and enhanced oxygen transfer throughout the culture medium. Fabricated from high-quality material, the bioreactor incorporates multiple process ports, sampling and addition vessels, and drain valves for convenient operation. It is suitable for microbial fermentation, enzyme production, biotechnology research, process optimization, and pilot-scale bioprocess development.

- ***Available in various capacities with custom configurations***

Technical features

Volume	
Stir Speed	
Stirring Mechanism	**Mechanical Agitation with Top-Drive Motor**
Body and Lid Material	
Weight	

Photochemical Products

GMC Technology

Triple-Jacketed Photochemical Reactor

The **Triple-Jacketed Photochemical Reactor** is designed for laboratory-scale photochemical and photocatalytic reactions under controlled irradiation conditions. The assembly comprises an inner quartz immersion well for housing the Xenon lamp, a middle cooling jacket for efficient water circulation, and an outer borosilicate glass reaction vessel for the reaction medium.

- Inner tube – houses the lamp.
- Middle water jacket – circulates coolant to control lamp temperature.
- Outer borosilicate reaction chamber - contains the reaction mixture.
- Compatible with UV, visible, and xenon light sources

Available in various capacities with custom configurations

Technical features

Material Vessel	
Material Lid	
Reactor Design	**Threaded Outer Borosilicate reactor with Teflon**
Reactor Bottom	**Flat suitable for magnetic stirring**

https://techinstro.com/photochemical-reactor/?utm

https://www.dolphinlabware.com/jacketed-reactor.html

4-Layer Jacketed UV Photochemical Reactor

The **4-Layer UV Lamp Reactor** is designed for laboratory-scale photochemical and photocatalytic reactions requiring uniform UV irradiation and efficient temperature control. The reactor consists of a central quartz immersion well for housing the UV lamp, surrounded by concentric cooling and reaction chambers to maximize light utilization while maintaining stable operating conditions. The upper assembly is fitted with a **PTFE cover**, providing excellent chemical resistance, leak-proof sealing, and secure mounting of the immersion well and process connections. Coolant is circulated through the jacket to maintain the lamp temperature, while the outer borosilicate glass vessel contains the reaction mixture. A **gas sparging tube** located at the bottom of the reactor allows the introduction of **nitrogen or other inert gases** for oxygen-free reactions, continuous purging, or gas-liquid photochemical processes.

- Compatible with UV, visible, and xenon light sources

Available in various capacities with custom configurations

Technical features

Capacity	**3L**
Reaction Vessel	**20 L Borosilicate Glass 3.3 with multiport lid**
Material of Construction (Lid)	
Operating Temperature/Design Temperature	**5°C/20°C**
Immersion Well	**Double Jacketed Quartz**
Pressure	**Atmospheric Pressure**
Cooling Jacket	**Independent Jacket**

4-Layer Jacketed Photochemical Reactor

The **4-Layer UV Lamp Reactor (20 L)** is designed for pilot-scale and industrial photochemical processes requiring efficient UV irradiation, precise temperature control, and continuous operation. The reactor features a **central quartz immersion well** housing the UV lamp, surrounded by concentric cooling and reaction chambers to maximize light utilization and ensure uniform irradiation of the reaction medium. The upper assembly is fitted with a **PTFE top cover**, providing excellent chemical resistance, leak-proof sealing, and secure mounting of the immersion well and process connections. The reactor incorporates a **bottom gas sparger** for nitrogen or other inert gas circulation, enabling oxygen-free reaction conditions and efficient gas–liquid contact. The complete assembly is mounted on a **heavy-duty stainless steel support framework**, ensuring excellent mechanical stability, ease of installation, and convenient operation for pilot-scale and industrial applications.

- ***Available in various capacities with custom configurations***

Technical features

Reaction Vessel	20 L Borosilicate Glass 3.3 with multiport lid
Immersion Well	Double Jacketed Quartz
Cooling Jacket	Independent Jacket
Support Structure	SS304 Framework
Operating Temperature/Design Temperature	5°C/20°C
Operating Pressure	Atmospheric Pressure

5-Layer Jacketed Photochemical Reactor

The 5-Layer Photochemical Reactor (1.5 L) is designed for laboratory-scale photochemical and photocatalytic applications requiring efficient UV irradiation and precise temperature control. The reactor features a **central quartz immersion well** for housing the UV lamp, surrounded by **multiple concentric cooling jackets** that ensure effective heat dissipation and uniform irradiation of the reaction medium. Separate inlet and outlet connections provide efficient cooling water circulation, while an additional port allows **nitrogen or other inert gas purging** for oxygen-free reaction conditions.

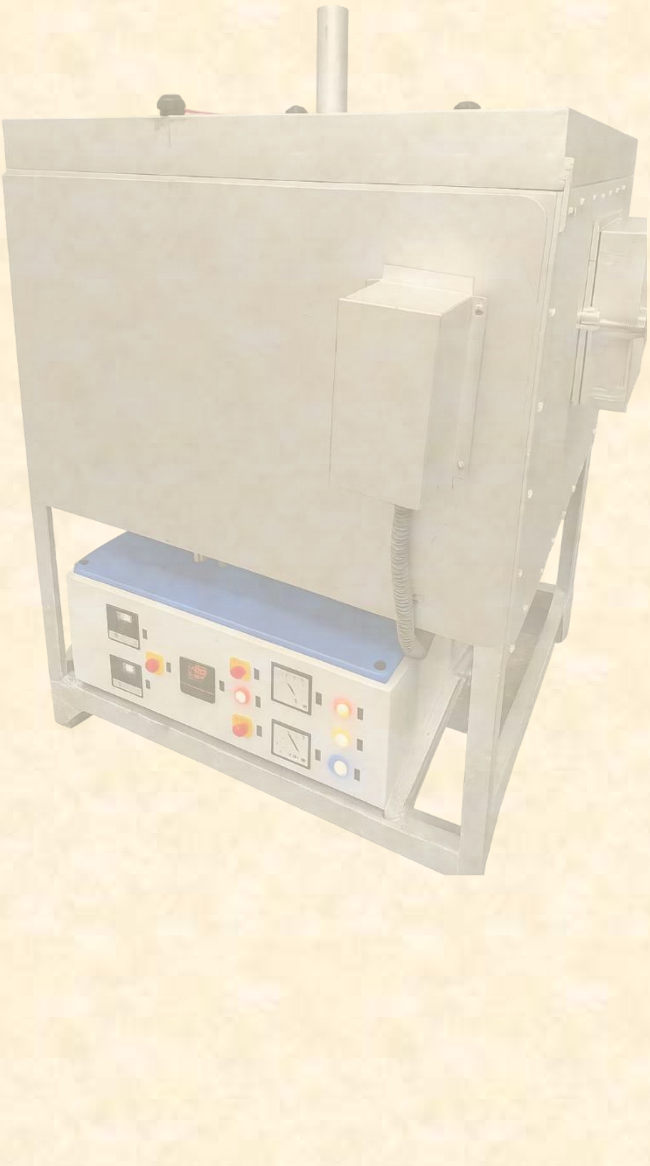
- Compatible with UV, visible, and xenon light sources

Available in various capacities with custom configurations

Technical features

Reaction Vessel	
Immersion well	
Operating Temperature/Design Temperature	
Operating Pressure	
Port	**Customization of port is available**

Special Instrumental Work

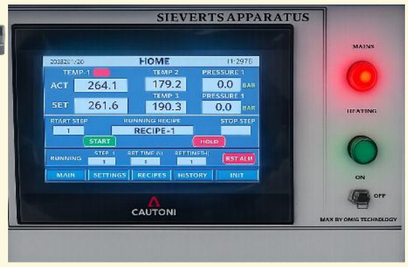


GMC Technology

High Pressure Hydrogen Distribution System

Special Instrument Work

Complete Assembly with Distribution System and Control Display Panel



The **High-Pressure Hydrogen Gas Distribution System** is a robust and precision-engineered manifold designed to provide a safe, reliable, and continuous supply of hydrogen gas to laboratory and pilot-scale process equipment. Fabricated from high-quality stainless steel tubing, fittings, and valves, the system offers excellent mechanical strength, corrosion resistance, and leak-tight performance under high-pressure operating conditions. The distribution panel is equipped with precision pressure gauges, isolation valves, and multiple outlet connections for accurate pressure monitoring and controlled gas delivery to one or more reactors simultaneously.

- » **Inlet Pressure Gauges:** 2 × 0–210 kg/cm²
- » **Outlet Pressure Gauge:** 1 × 0–14 kg/cm² (customizable)
- » **Flow Rate:** Up to 300 LPH
- » **Inlet Connection:** ¼" BSP male to suit cylinder pigtails
- » **Outlet Connection:** ¼" OD ferrule with precision ball valve
- » **Mounting Dimensions:** 500 mm (L) × 300 mm (H) × 45 mm (D)
- » **Construction Materials:** Brass (chrome-plated) or SS316 stainless steel

Technical features

Material	
Inlet Pressure/Outlet Pressure	
Flow Rate	
Mounting Dimension	
Inlet Connection	
Outlet Connection	

https://athenatechnology.in/product/gas-manifold-system/?utm

Polarimeter Tube

Special Instrument Work

The body of a polarimeter tube is not required to be transparent, as the optical path is confined to the longitudinal axis of the tube. During measurement, polarized light passes only through the front quartz (or optical glass) window, the liquid sample, and the rear quartz (or optical glass) window, while the side wall does not contribute to the optical path. The transparent body is primarily provided to facilitate visual inspection of the sample, including monitoring liquid level, bubble formation, or contamination. Consequently, **a stainless steel (SS)** tube fitted with precision quartz windows at both ends is a suitable and widely accepted design, offering superior mechanical strength, chemical resistance, and durability without affecting optical performance.

- ***Precision engineered***
- ***Available in custom configurations***

Technical features

Material (inside tube and outside)	
Length/internal Diameter	
volume	
Seal (Tube/end cap)	

https://www.xylemanalytics.com/en/glass-polarimeter-tube-centre-metal-ends

Gas Pressure Regulator

Special Instrument Work

The **High-Pressure Gas Regulator** is designed to accurately reduce and control the pressure of gases supplied from high-pressure cylinders for laboratory and industrial applications. The regulator ensures a stable and consistent gas outlet pressure while enabling safe operation of downstream equipment. Equipped with precision pressure gauges and a robust control mechanism, it allows continuous monitoring of both cylinder and delivery pressures. Manufactured from high-quality stainless steel, the regulator offers excellent corrosion resistance, leak-tight performance, and long service life. It is suitable for use with hydrogen, nitrogen, helium, argon, and other industrial or specialty gases.

Technical features

Pressure Rating (Inlet/Outlet)	
Material	
Hose pipe dimension	

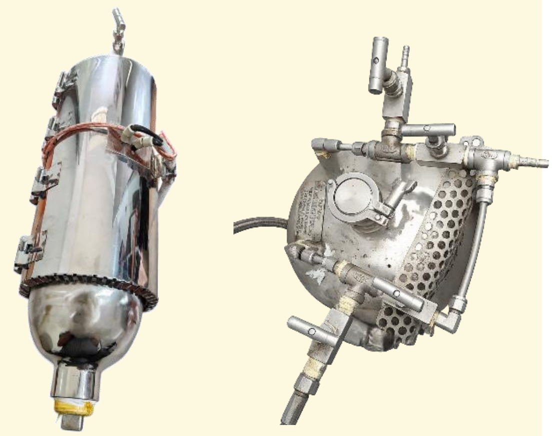
https://amarequip.com/accessories-models/forward-pressure-regulator

https://discreteautomation.emerson.com/product/tescom-high-pressure-panel?utm

Solvent Purification System

Special Instrument Work

Description of solvent purification system



Technical features

Muffle Furnace (long tunnel 1.5 Metre)

Special Instrument Work

The **Long Tunnel Furnace** is designed for continuous thermal processing of materials under controlled heating conditions. The furnace comprises a robust outer chamber fabricated from heavy-duty steel with a high-temperature resistant finish, while the inner heating chamber is lined with high-grade refractory insulation to ensure excellent thermal efficiency and minimal heat loss. The heating zone is insulated with ceramic fiber to provide uniform temperature distribution throughout the chamber. The furnace is equipped with a front-loading access door, digital temperature controller, and electrical control panel for precise temperature regulation and safe operation.

- ***Available in various capacities with custom configurations***

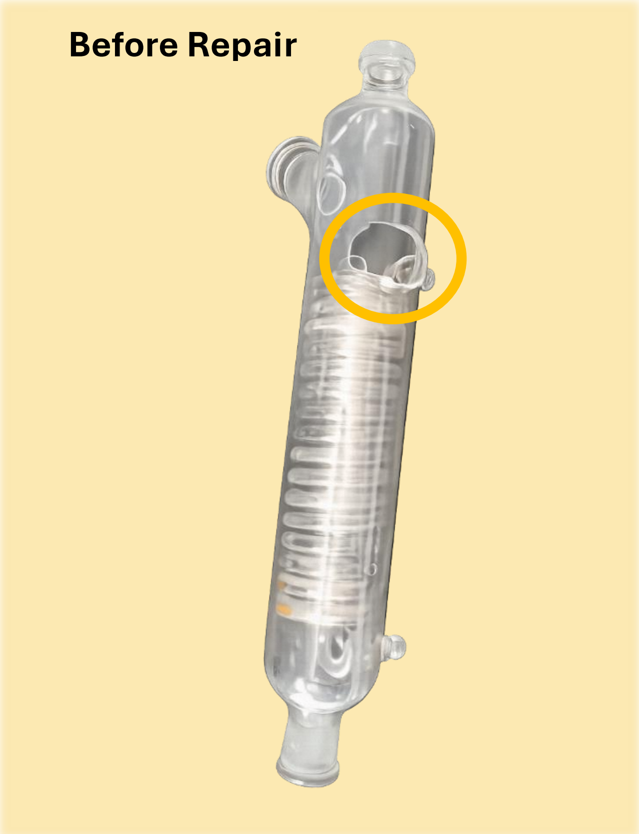
Technical features

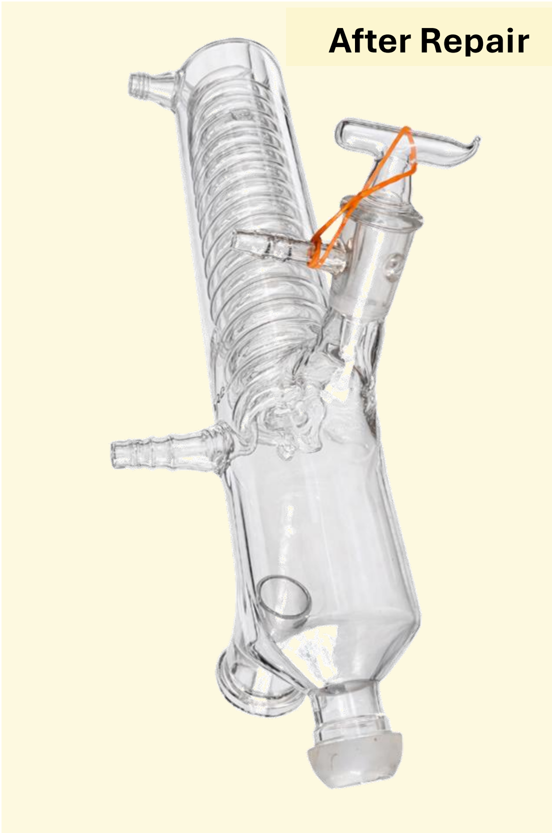
Operating Temperature	
Maximum Temperature	**800ºC**
Temperature Accuracy	
Power supply	
Material and Insulation	

https://www.digiqualsystems.com/products/furnaces/muffle-furnaces/900c/?utm_

Rotary Evaporator Condenser

Special Instrument Repair Work





We undertake **professional repair** and **refurbishment of Rotary Evaporator glassware**, restoring damaged laboratory glass components through precision glassblowing. Broken or cracked components are carefully reconstructed while maintaining the original dimensions, alignment, and functionality. Each repaired assembly is inspected for structural integrity, leak-tight performance, and reliable operation, offering a cost-effective alternative to replacement.

Customized-Glassware Work

GMC Technology

Customized DEO Assembly

The advanced DEO System is designed for efficient deodorization, purification, and treatment of industrial oils. The system effectively removes odor, moisture, volatile impurities, and dissolved gases to improve oil quality and performance. Suitable for edible oils, transformer oils, lubricants, and specialty oil applications.

- Applications: Edible oil processing, transformer oil treatment, lubricating oil purification, refinery and chemical process industries.
- Optional Features: Vacuum system, condenser unit, PLC automation, skid-mounted design.

Technical features

Material	
Temperature	
Size/Dimension	**As per requirement**
Joints	**As per requirement**

https://www.dolphinlabware.com/essential-oil-steam-distillation-unit.html

Customized Special glassware

4-Neck cylindrical flask with accessories

The **Reaction Flask** Assembly is designed for laboratory-scale synthesis, reflux, and distillation applications. The assembly comprises a multi-neck borosilicate glass reaction flask equipped with interchangeable water-cooled condensers, enabling flexible configurations to meet different experimental requirements.

- ***Available in various capacities with custom configurations***

Solvent Still Head

The **Solvent Distillation Head** is designed for the safe distillation, collection, and controlled transfer of solvents under laboratory conditions. Fabricated from Borosilicate Glass 3.3, the assembly incorporates standard ground-glass joints and PTFE stopcocks for controlled solvent dispensing and efficient operation.

- ***Available in various capacities with custom configurations***

Customized Special glassware

Glass Reflex Divider is manufactured from high-quality borosilicate/reflex glass for laboratory, chemical, and industrial process applications requiring visual observation, thermal resistance, and chemical compatibility. Suitable for reactors, UV chambers, sight assemblies, and vacuum systems.

Applications: Laboratory equipment, chemical process systems, UV exposure units, reactors, and sight observation assemblies.

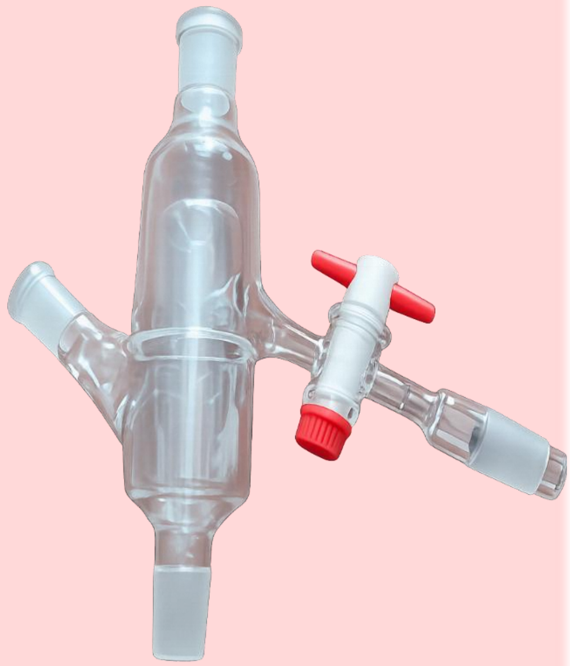
Technical Features:

Material	**Borosilicate glass 3.3**
Joints	**Custom**
Valve	**PTFE, lateral, adjustable**
Size/Dimension	**As per requirement**
Product Form	**Tube**

https://www.rettberg-labstore.com/en/products/distillation-reflux-divider-manually-controlled?utm

https://www.sabarscientific.net/glass-reflux-divider.html?utm

Customized Pressure Equalizing Adapter



The **Multi-Port Reaction Adapter** is designed for controlled reagent addition, gas introduction, vacuum operation, and sampling in laboratory reaction systems. The assembly incorporates standard taper ground-glass joints and an integrated PTFE stopcock for precise flow control. It is suitable for use with reaction kettles, photochemical reactors, and inert atmosphere reaction assemblies.

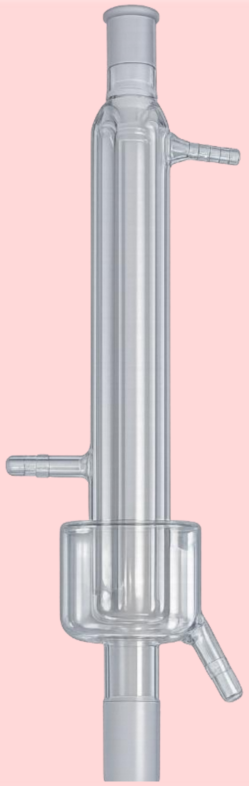
Specification

Top angled female ground-glass joint-for connecting an addition funnel, condenser, thermometer adapter, or gas inlet.
Horizontal main body with male and female ground-glass joints-connects between the reaction vessel and the upper glassware.
Central pressure-equalizing chamber-helps maintain pressure equilibrium during reagent addition.
Bottom male ground-glass joint-fits into the reactor or reaction kettle cover.

Technical Features

Material	**Borosilicate glass 3.3**
Joints	**As per requirement**
Capacity	**50 ml, 100ml**
Size/Dimension	**As per requirement**

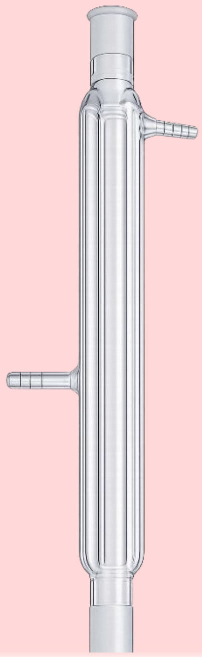
Customized Double-Jacketed Condenser



Double-jacketed Condenser with Receiver Cup

The **double-jacketed condenser with reservoir cup** is designed for efficient condensation of solvent vapours during reflux and distillation operations. The condenser is manufactured from Borosilicate Glass and fitted with standard taper ground-glass joints for leak-tight connection with reaction assemblies.

Want to add more features about resirvoir…..



Double-Jacketed Condenser

The **Double-Jacketed Condenser** is designed for efficient condensation of solvent vapours during reflux and distillation processes. Manufactured from **Borosilicate Glass 3.3**, it features a double-jacketed cooling chamber that provides effective heat transfer and reliable solvent recovery.

Customized Aluminium and Ceramic Wool-Insulated Vacuum Column

Specifications

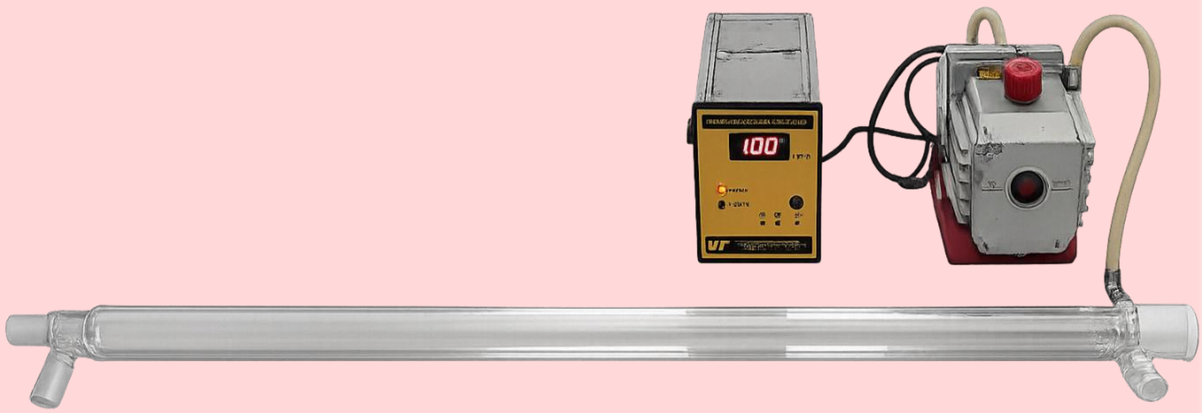
Customized General Glassware with Specifications for Vacuum System

Large size vacuum trap

Specifications

Customized Vacuum Insulated 1200 mm Column

Vacuum Insulated 1200 mm Column



Specifications

Customized Reaction Kettle Assembly

The borosilicate glass reaction kettle assemblies are designed for laboratory-scale chemical synthesis, photochemical reactions, catalysis, and process development. The modular construction comprises interchangeable kettle bodies and multi-neck kettle covers, allowing flexible installation of condensers, mechanical stirrers, thermometers, gas inlet/outlet adapters, and addition funnels. The assemblies are secured using **stainless steel quick-release clamps** and **PTFE sealing gaskets** to ensure a leak-tight seal during operation. The reaction kettles are available in standard and bottom-discharge configurations to facilitate efficient charging, reaction, and product recovery.

Specifications

Material	
Capacity	
Pressure	
Size and Dimension	**As per requirement**
Mouth Style	**Wide**
Temperature	

https://www.fishersci.com/us/en/browse/90094053/reaction-kettles

Customized General Glassware

Schlenk tube

Sample/Storage bottle with glass joint

Glass crucible without lid

Specifications

The **Schlenk Tube** is designed for handling air- and moisture-sensitive compounds under vacuum or inert atmosphere. Manufactured from Borosilicate Glass 3.3, the assembly incorporates a PTFE high-vacuum valve and a sidearm for gas or vacuum connection.

Size and Dimension: As per requirement

The 20 mL **Glass Storage Bottle** with Ground Joint is designed for the safe storage and handling of laboratory reagents, reference standards, and moisture-sensitive samples. Manufactured from Borosilicate Glass 3.3, it features a precision-ground glass joint for a secure, leak-resistant seal.

Size and Dimension: As per requirement

Manufactured from Borosilicate Glass 3.3, this **crucible** is suitable for general laboratory applications involving heating, drying, and sample handling. It offers good resistance to thermal shock and chemical attack.

Size and Dimension: As per requirement